

INVESTMENTS

Ninth Canadian Edition · Bodie, Kane, Marcus

CHAPTER 13

Empirical Evidence on Security Returns

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

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Study Guide · Prepared for Personal Use

LO1 · The Index Model and the Single-Factor APT

This chapter surveys how researchers have actually **tested** the CAPM and APT models developed in Chapters 9–10. The most commonly tested implication is the expected return–beta relationship, tested via a standardized **two-pass regression** methodology.

The Two-Pass Regression Methodology

- **First-pass regression (estimating the SCL)** — for each of many stocks (e.g., 100), regress 60 months of excess returns on the market's excess returns, exactly as in Chapter 8. This yields, for every stock: an estimated beta, an average excess return, and an estimate of residual (firm-specific) variance.
- **Second-pass regression (estimating the SML)** — using the betas estimated in the first pass as the independent variable, regress the sample of stocks' average excess returns against their betas. If the CAPM holds exactly, the intercept (γ_0) should be zero and the slope (γ_1) should equal the average market risk premium.

An expanded second-pass regression can also include each stock's residual variance, $\sigma^2(e_i)$, as an additional independent variable. The CAPM predicts this coefficient (γ_2) should be zero — firm-specific risk should carry no risk premium, since it can be diversified away.

A key early methodological improvement (Black, Jensen, and Scholes; later refined by Fama and MacBeth) was to group stocks into diversified portfolios (rather than testing individual stocks) before running the analysis, since individual stock betas are estimated with substantial error, while portfolio-level betas average out much of that estimation noise.

LO2 · Tests of the Multi-Factor CAPM and APT

Fama and MacBeth (1973) applied the portfolio-based two-pass methodology across many periods (rolling 5-year estimation windows) and famously found the empirical SML to be **flatter than the CAPM predicts**: the intercept (γ_0) was **positive** rather than zero, and the slope (γ_1) was **less than** the average market risk premium. In practice, this means **low-beta stocks earned higher average returns than the CAPM predicted, while high-beta stocks earned lower returns than predicted**. A further troubling finding: the coefficient on residual (firm-specific) risk, γ_2 , was often **positive and economically meaningful** — directly contradicting the CAPM's prediction that only systematic risk should be priced.

Multifactor Extensions

Because a single-factor model appears to leave systematic patterns unexplained, researchers extended tests to multiple factors (paralleling Chapter 10's multifactor APT): adding factors for firm size, momentum, and other characteristics. These multifactor tests generally show improved explanatory power over the single-factor CAPM, but reintroduce the **joint hypothesis problem** from Chapter 11 in sharper form: a poor fit could mean the pricing model is wrong, or that the *proxy* used for the true market/factor portfolio is flawed (Roll's critique). Richard Roll specifically argued that since the true, all-encompassing market portfolio is unobservable, apparent CAPM failures might simply reflect using a poor stand-in benchmark (e.g., the S&P 500) rather than a genuine failure of the underlying theory.

LO3 · Fama-French-Type Factor Models

Building on Chapter 10's Fama-French three-factor model (market, SMB, HML), researchers test whether these firm-characteristic-based factors actually earn statistically and economically significant risk premiums, and whether including them eliminates the CAPM's empirical anomalies (the size effect, the value/book-to-market effect).

The evidence broadly supports SMB and HML as capturing **real, persistent** patterns in average returns beyond what market beta alone explains. But as flagged in Chapter 10, the interpretation remains contested: are SMB and HML proxies for *priced risk factors* (e.g., distress risk that the CAPM's single market factor misses), or do they simply reflect *mispricing* that a behavioural, non-risk-based story would explain instead? The factors' empirical power to explain average returns is not, by itself, evidence for either interpretation — both would produce the same statistical pattern.

Momentum (Chapter 11) is often added as an additional factor alongside market, SMB, and HML, forming a four-factor model. Even this expanded model leaves some patterns in average returns unexplained, motivating continued research into additional candidate factors (e.g., profitability, investment intensity).

LO4 · Liquidity and Asset Pricing

A more recent and increasingly influential extension incorporates **liquidity** — the ease and cost of converting an asset to cash without materially moving its price — directly into asset pricing. Illiquid securities should trade at a discount (offer a higher expected return) to compensate investors for the cost and difficulty of trading them.

Measuring Illiquidity

Pastor and Stambaugh study **return reversals** associated with high trading volume: if part of an observed price move reflects a temporary liquidity concession (e.g., a large seller having to accept a lower price to complete a trade quickly) rather than genuine new information, that portion should partially reverse afterward. They find reversals are indeed systematically larger following higher-volume trades — direct evidence of a liquidity-driven price component.

Amihud proposed a widely used illiquidity measure: the monthly average of a stock's daily **|return| ÷ dollar trading volume** (see Formulas page) — capturing the price impact per dollar of trading. Higher values indicate a given dollar volume of trading moves the price more, signaling greater illiquidity. Switzer and Picard show that a broad worsening in aggregate market liquidity (via the Amihud measure) tends to **precede future contractions in real GDP growth**, suggesting liquidity conditions carry genuine macroeconomic, business-cycle-relevant information, not just microstructure noise.

Acharya and Pedersen extend this further into a full **liquidity-adjusted CAPM**, in which a security's required return depends not only on its return's covariance with the market, but also on the covariance of its *liquidity* with the market's overall liquidity — since an asset that becomes especially illiquid exactly when the whole market is under liquidity stress is particularly unattractive to hold.

LO5 · Consumption-Based Asset Pricing and the Equity Premium Puzzle

The **Consumption-based CAPM (CCAPM)** reframes risk fundamentally: what ultimately matters to an investor is not wealth for its own sake, but the **lifetime flow of consumption** that wealth can support. Under this lens, a security's risk premium should depend on the covariance of its return with growth in **aggregate consumption** (via a consumption-tracking portfolio), rather than simply its covariance with the market portfolio's return (see Formulas page).

The Equity Premium Puzzle

Mehra and Prescott (1985) observed that historical U.S. excess stock returns are **far too large** to be justified by the CCAPM given any *reasonable* level of investor risk aversion — the implied covariance between consumption growth and market returns is simply too low relative to the risk premium actually observed historically. This mismatch is the **equity premium puzzle**, and it has also been documented in Canadian data: over 1956–2006, the average equity risk premium was approximately **4.2% for Canada versus 5.9% for the U.S.** — both substantial, and both larger than simple consumption-based models can readily explain.

Proposed Resolutions

- **Consumption-tracking portfolios** — rather than using noisy, infrequent raw consumption data directly, researchers (e.g., Jagannathan and Wang) construct portfolios engineered to closely track consumption growth, improving statistical power — though any imprecision in this tracking construction still muddies the results.
- **Sample-period sensitivity** — Fama and French argue the historically observed equity premium is partly a "creature of modern times" — unusually high in the specific post-war U.S. sample most often studied, and less puzzling when a longer or different historical window is used.
- **The illiquidity premium** — some of the "excess" historical equity return may actually be fair compensation for equities' relative illiquidity (LO4) rather than a pure risk-aversion puzzle at all — potentially of a similar order of magnitude to the unexplained premium itself.

The puzzle carries a genuinely important practical implication: if the historical U.S./Canadian equity premium substantially overstates the *truly required* compensation for risk, then forward-looking return forecasts based on simple historical averages may be too optimistic — a caution echoed from Chapter 5's discussion of historical risk premiums.

Chapter 13 — Big-Picture Takeaways

- The standard test of the CAPM/APT uses a two-pass regression: first estimate each security's beta (SCL), then regress average returns on those betas (SML).
- Fama-MacBeth-style tests find a flatter-than-predicted SML (positive intercept, shallow slope) and a positive, significant coefficient on residual risk — both contradicting the simple CAPM.

- Roll's critique means poor test results can reflect either a genuinely wrong model or simply an inadequate proxy for the true, all-encompassing market portfolio.
- Fama-French-type factors (SMB, HML, often plus momentum) improve empirical fit but their interpretation as genuine priced risk versus mispricing remains actively debated.
- Liquidity is an increasingly important, empirically supported asset-pricing factor: illiquid securities earn a return premium, and aggregate liquidity conditions carry real macroeconomic signal.
- The equity premium puzzle — observed in both U.S. (5.9%) and Canadian (4.2%) data — shows historical stock risk premiums are difficult to reconcile with reasonable risk aversion under consumption-based asset pricing, with liquidity and sample-period effects among the leading proposed resolutions.

This chapter is primarily about interpreting empirical tests rather than performing new calculations, but these relationships anchor the discussion.

F1 | First-Pass Regression (Security Characteristic Line)

$$R_{it} = \alpha_i + \beta_i \times R_{Mt} + e_{it} \quad (\text{excess returns; } i = \text{stock, } t = \text{time period})$$

Estimated separately for each stock over the sample period, exactly as in Chapter 8, to obtain each stock's beta.

F2 | Second-Pass Regression (Security Market Line Test)

$$\bar{r}_i - \bar{r}_f = \gamma_0 + \gamma_1 \times b_i + \gamma_2 \times \sigma^2(e_i) + u_i$$

b_i = first-pass beta estimate; CAPM predicts $\gamma_0 = 0$, γ_1 = average market risk premium, $\gamma_2 = 0$. Empirically, $\gamma_0 > 0$, γ_1 too small, and γ_2 often significantly positive.

F3 | Amihud Illiquidity Measure

$$ILLIQ = \text{Monthly Average of } [| \text{Daily Stock Return} | / \text{Daily Dollar Volume}]$$

Higher ILLIQ means a given dollar amount of trading moves the price more — i.e., the stock is less liquid. Used to estimate both the liquidity cost and liquidity risk of a security.

F4 | Consumption-Based CAPM (CCAPM) Risk Premium

$$E(R_M) = A \times \text{Cov}(r_M, r_C)$$

A = coefficient reflecting average risk aversion; r_C = return on a consumption-tracking portfolio. The equity premium puzzle: the actual historical $E(R_M)$ is far larger than reasonable A and observed $\text{Cov}(r_M, r_C)$ can jointly justify.

EXAMPLE | Interpreting a Fama-MacBeth Second-Pass Regression

Given: For the 1946–1955 period, the risk-free rate averaged 8 basis points (0.08%) per month, and the average market risk premium was 103 basis points (1.03%) per month.

CAPM prediction for the second-pass regression:

γ_0 (predicted) = r_f = 0.08% per month

γ_1 (predicted) = market risk premium = 1.03% per month

γ_2 (predicted) = 0 (no reward for diversifiable risk)

If an actual Fama-MacBeth regression instead estimated γ_0 noticeably above 0.08% and γ_1 noticeably below 1.03%, this is the classic "flatter than predicted SML" finding: low-beta stocks earned more, and high-beta stocks earned less, than the CAPM predicts. If γ_2 also came back significantly positive, that would further suggest firm-specific risk was earning compensation — a direct contradiction of the CAPM.

EXAMPLE | The Equity Premium Puzzle — Canada vs. U.S.

Given: Switzer (2007) estimates the average annual equity risk premium over 1956–2006 was 4.2% for Canada and 5.9% for the U.S.

Interpretation: Under the CCAPM, both premiums would require investors to have a specific, reasonable level of risk aversion (A) combined with an observed covariance between consumption growth and market returns. Mehra and Prescott showed that for the U.S., no historically reasonable value of A can reconcile observed consumption covariance with a premium anywhere near 5.9% — the puzzle. The lower 4.2% Canadian figure is somewhat less extreme, but still substantially exceeds what simple consumption-based models predict, confirming the puzzle is not simply a U.S.-specific statistical artifact.

PRACTICE PROBLEM | Nose Hill Capital Research

Scenario

You are a research associate at Nose Hill Capital Research in Calgary, tasked with re-examining the CAPM using a sample of 80 TSX-listed stocks over 60 months.

Part A — The Two-Pass Methodology

- (i) Briefly describe what data and regression you would run in the "first pass" for each of the 80 stocks.
- (ii) Briefly describe what data and regression you would run in the "second pass."
- (iii) According to the CAPM, what should the intercept and slope of the second-pass regression equal?

Part B — Interpreting Your Results

Your second-pass regression produces: $\gamma_0 = 0.35\%$ per month, $\gamma_1 = 0.60\%$ per month, $\gamma_2 = 0.18$ (significant).

The average monthly T-bill rate over your sample was 0.20%, and the average monthly market risk premium was 0.90%.

- (i) Compare your estimated γ_0 and γ_1 to their CAPM-predicted values. Is your SML flatter or steeper than the CAPM predicts?
- (ii) What does the significant, positive γ_2 suggest about the pricing of firm-specific risk in your sample?
- (iii) A colleague says these results definitively prove the CAPM is wrong. Using Roll's critique, explain why this conclusion may be too strong.

Part C — Liquidity

Nose Hill wants to compute the Amihud illiquidity measure for two stocks: Bow Ridge Corp (average daily |return| = 1.2%, average daily dollar volume = \$2 million) and Foothills Energy (average daily |return| = 0.8%, average daily dollar volume = \$40 million).

- (i) Calculate each stock's Amihud illiquidity measure.
- (ii) Which stock is more illiquid, and what does this predict about its required return relative to the other, all else equal?

Part D — The Equity Premium Puzzle

A client asks Nose Hill to justify using a 6% annual equity risk premium in a long-term financial plan, citing "historical Canadian and U.S. averages."

- (i) Referencing LO5, explain one reason this historical average might overstate the true risk premium investors should require going forward.
- (ii) Name two of the proposed resolutions to the equity premium puzzle discussed in this chapter.

Hints & Guidance

Part A tests LO1 and Formulas F1–F2. Part B tests LO1–LO2 (mirror the Fama-MacBeth example's logic with your own numbers). Part C tests LO4 and Formula F3. Part D tests LO5 directly.